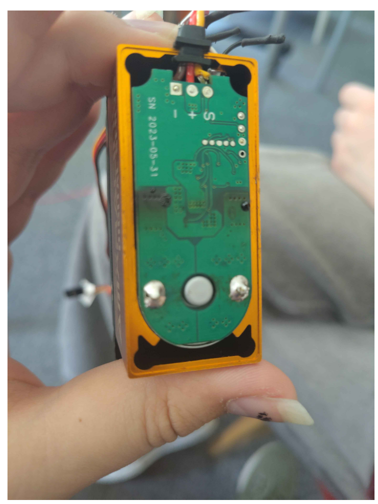
This document will note and explain the changes done to the LIMB project during the summer of 2026

1 Mechanical

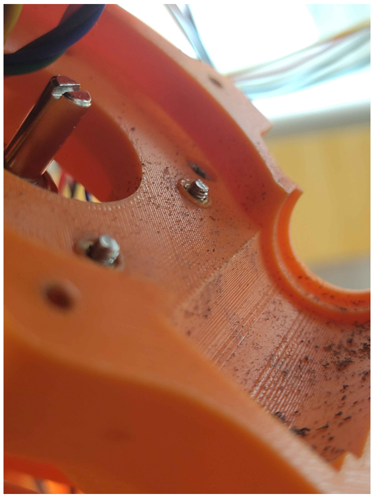
1.1 Shoulder Up-Down Worm & Worm gear

The original worm broke.

A new worm was printed out of carbonfibre. It also broke along with the motor driving it.



The reason for both these failures were the length of screws connecting the potentiometer being too long and digging into the worm gear.



This caused enough friction to require the motor to work incredibly hard until the breaking of the part gave the motor a back current strong enough to fry the MOSFETs.

The screws were exchanged for shorter ones, the gear, worm and casing were printed anew.

1.2 Outer Bicep

Underneath the outercasing of the bicep there is a step down circuit which was not acounted for in the original design which made the plate crack when tightened.

The new plate is thickened to enable a dugout for this part while still fitting the surounding parts.

The skeletal board underneath ought to be extended to match this new hight to cover the dugout from view.

1.3 Inner Bicep

Simply made to match the outer Bicep for aesthetic reasons.

1.4 Finger Nylon Thread

The nylon thread connecting the ring finger was replaced. The underlying sharp edges were not addressed.

2 Electrical

2.1 Powerline Quick Disconnect

2.2 Shoulder Rotation Quick Disconnect

2.3 Upper Arm IMU Cable Connector

3 Software